

DATABASE MANAGEMENT SYSTEMS (CS301)

Concurrency Control Basics

Learning Objectives

Understand concurrency control, transaction conflicts, locking mechanisms, deadlocks, and recovery concepts in DBMS.

Introduction to Concurrency Control

Concurrency control is the process of managing simultaneous execution of transactions without compromising database consistency and integrity.

Need for Concurrency Control

- Supports multiple users simultaneously
- Prevents data inconsistency
- Ensures transaction isolation
- Maintains database correctness

Lost Update Problem

Occurs when two transactions update the same data item and one update overwrites the other, causing loss of information.

Dirty Read Problem

Occurs when a transaction reads data that has been modified by another transaction but not yet committed.

Unrepeatable Read

Occurs when the same query returns different results because another transaction modifies the data between reads.

Lock-Based Protocols

Locks are used to control concurrent access to database items and prevent conflicts among transactions.

Shared Lock (S-Lock)

Allows multiple transactions to read the same data item but prevents modifications.

Exclusive Lock (X-Lock)

Allows a transaction to read and modify a data item while preventing access by other transactions.

Two-Phase Locking (2PL)

Growing Phase: Locks can be acquired but not released.

Shrinking Phase: Locks can be released but not acquired.

Deadlocks

A deadlock occurs when two or more transactions wait indefinitely for resources held by each other.

Deadlock Prevention

- Resource ordering
- Timeout mechanisms
- Preemption techniques

Recovery Concepts

- Log-based Recovery
- Checkpoints
- Rollback Operations
- Crash Recovery

Applications

- Banking Systems
- Online Shopping Platforms
- Airline Reservation Systems
- Hospital Information Systems

Exam Important Questions

1. Define concurrency control.
2. Explain lost update and dirty read problems.
3. Differentiate shared and exclusive locks.
4. Explain Two-Phase Locking (2PL).
5. What is a deadlock?
6. Discuss recovery techniques in DBMS.

Summary

Concurrency control ensures safe and efficient execution of multiple transactions. Locking protocols, deadlock handling, and recovery mechanisms are essential for maintaining database reliability.